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Oefeningenreeks 4A nr. 15.3

$$\begin{aligned} & \int x^3 e^{3x} dx \\ & \left\{ \begin{array}{l} u = x^2 \\ dv = e^{3x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2x dx \\ v = \frac{e^{3x}}{3} \end{array} \right. \\ & = \frac{x^2 e^{3x}}{3} - \frac{1}{3} \int 2x e^{3x} dx \\ & \left\{ \begin{array}{l} u = 2x \\ dv = e^{3x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2 dx \\ v = \frac{e^{3x}}{3} \end{array} \right. \\ & = \frac{x^2 e^{3x}}{3} - \frac{2x e^{3x}}{9} - \frac{1}{9} \int e^{3x} 2 dx \\ & = \frac{x^2 e^{3x}}{3} - \frac{2x e^{3x}}{9} + \frac{2e^{3x}}{27} + C \end{aligned}$$

References